Ali Awais Safdar

BS Computer Science | National University of Sciences and Technology (NUST)
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PROFESSIONAL PROFILE

Aspiring computer scientist with a strong foundation in AI, machine learning, and software development, seeking to leverage my expertise in advanced technologies like NLP, deep learning, and blockchain to create innovative and impactful solutions. With hands-on experience in developing medical diagnostic tools, resilient software architectures, and multilingual AI models, I aim to contribute to cutting-edge projects that bridge the gap between research and real-world applications. Passionate about driving technological advancements and fostering inclusive innovation, I am committed to utilizing my skills to address complex challenges and deliver meaningful outcomes in a dynamic and collaborative environment.

EDUCATION

- BS in Computer Science School of Electrical Engineering and Computer Science , 2.96
- Bachelors in Computer Science School of Electrical Engineering and Computer Science , (2025)
- Pre Engineering Punjab College , (2021)
- Matriculation - Biology Garrison Boys High School , (2019)

WORK & INTERNSHIP EXPERIENCE

Nanyang Technological University, Singapore
01-Jul-2024 - 16-Aug-2024

Vision Architech (Remote)
08-Jul-2024 - 09-Sep-2024

HPC Lab-SEECS
25-Dec-2023 - 27-Dec-2024

CarbonTeq
03-Jun-2024 - 05-Jul-2024

Software Productivity Strategists, Inc. (Remote)
06-Nov-2023 - 05-Feb-2024

Radar Research Lab, SINES
02-Oct-2023 - 08-Jan-2024

PROJECTS

Enhancing Vehicle Routing with Time Windows: A Machine Learning-Driven Ant Colony Optimization Approach

- Authored a research paper focusing on integrating machine learning techniques with Ant Colony Optimization (ACO) to improve vehicle routing under time-window constraints.
- Presented the paper at the FIT 2024 Conference, showcasing innovative approaches to solving logistics and optimization challenges using AI

PATHVQA-VGG19-BERT-ATTENTION

- Developed a medical Visual Question Answering (VQA) system utilizing VGG-19 for image feature extraction and BERT for textual question processing.

- Incorporated attention mechanisms to enhance contextual understanding, facilitating accurate responses to complex medical queries.
- Created an interactive web int

Secure Information Handling with Blockchain

- Implemented secure data handling solutions using blockchain technology with cryptographic methods such as HMAC-SHA256 for hashing and RSA for encryption.
- Developed a decentralized ledger to ensure tamper-resistance and transaction verification in distributed environments.

DNS Client Efficient Design

- Engineered a sophisticated TypeScript-based DNS client with a modular, scalable architecture and robust error handling.
- Optimized performance using asynchronous processing and railway-oriented programming principles.

TECHNICAL SKILLS

- Data engineering
- Blackbox/whitebox testing
- Laptop repair
- Agile and waterfall methods
- Distributed systems
- Python socket programming
- Cuda
- Big data analytics

SOFT SKILLS

- Financial ratio analysis
- Statistical data analysis
- Voracious reader
- Conflict management
- Team management skills
- Multi tasking
- Group work
- Communication oral/written